

✦ **AAA (Authentication, Authorization, Accounting):** Three core functions, AAA in Networking & Security, AAA Deployment Models..

[Q-1] Write a short scenario where a user tries to access a private WiFi network (like at a campus or airport) and describe how each of the three AAA functions—Authentication, Authorization, and Accounting—would apply step by step in this situation.

(A)

○ **Scenario**

Suppose a student wants to connect to a private WiFi network at a college campus. The WiFi is protected and only registered students are allowed to access it.

○ **Step-by-step AAA process**

1. Authentication – “Who are you?”

The student connects their laptop or mobile phone to the college WiFi and enters their student username and password. The WiFi access point sends these credentials to the AAA server, such as a RADIUS server. The server checks the credentials against the college's user database. If the credentials are correct, the student is authenticated.

2. Authorization – “What are you allowed to do?”

After successful authentication, the AAA server determines what the student is permitted to access. For example, the student may be allowed to access the Internet, college websites, learning platforms, and email, but may not be allowed to access administrative servers or restricted network resources.

3. Accounting – “What did you do?”

The AAA server records information about the student's network session. It may log the username, login time, logout time, IP address, and amount of network usage. These records can be used for security monitoring, troubleshooting, and auditing.

- Student → WiFi Access Point → AAA/RADIUS Server → Authentication → Authorization → Accounting/Logging

[Q-2] Create a table comparing at least two different AAA deployment models (such as centralized vs distributed), listing three advantages and three disadvantages of each for a company like Zomato that handles sensitive user data.

(A)

○ Comparison of AAA Deployment Models

Feature	Centralized AAA	Distributed AAA
Description	Authentication, authorization, and accounting are handled by one or more central AAA servers.	AAA services are deployed across multiple servers or network locations.
Advantage 1	Users and security policies are easier to manage from one location.	Provides better availability because multiple AAA servers can operate independently.
Advantage 2	Security policies can be applied consistently across the organization.	Can reduce network delay by placing AAA servers closer to users or branches.
Advantage 3	Centralized logging makes auditing and monitoring easier.	Provides better scalability for organizations with multiple locations.
Disadvantage 1	Failure of the central AAA server can affect many users.	Managing and maintaining multiple AAA systems is more difficult.
Disadvantage 2	A large number of requests can overload the central server.	Policies can become inconsistent if servers are not properly synchronized.
Disadvantage 3	Remote branches may depend on connectivity to the central AAA server.	Requires additional infrastructure and maintenance costs.

○ Example for Zomato

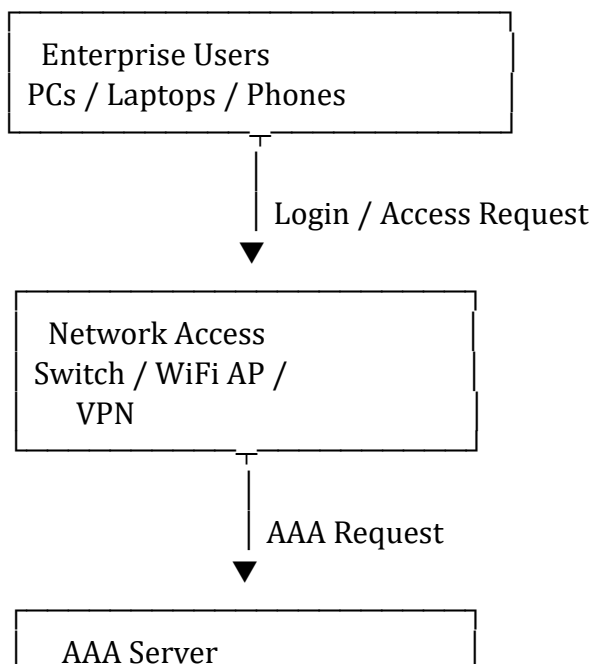
For a company such as Zomato, which handles sensitive customer, restaurant, employee, and payment-related information, AAA deployment should provide strong access control and reliable logging.

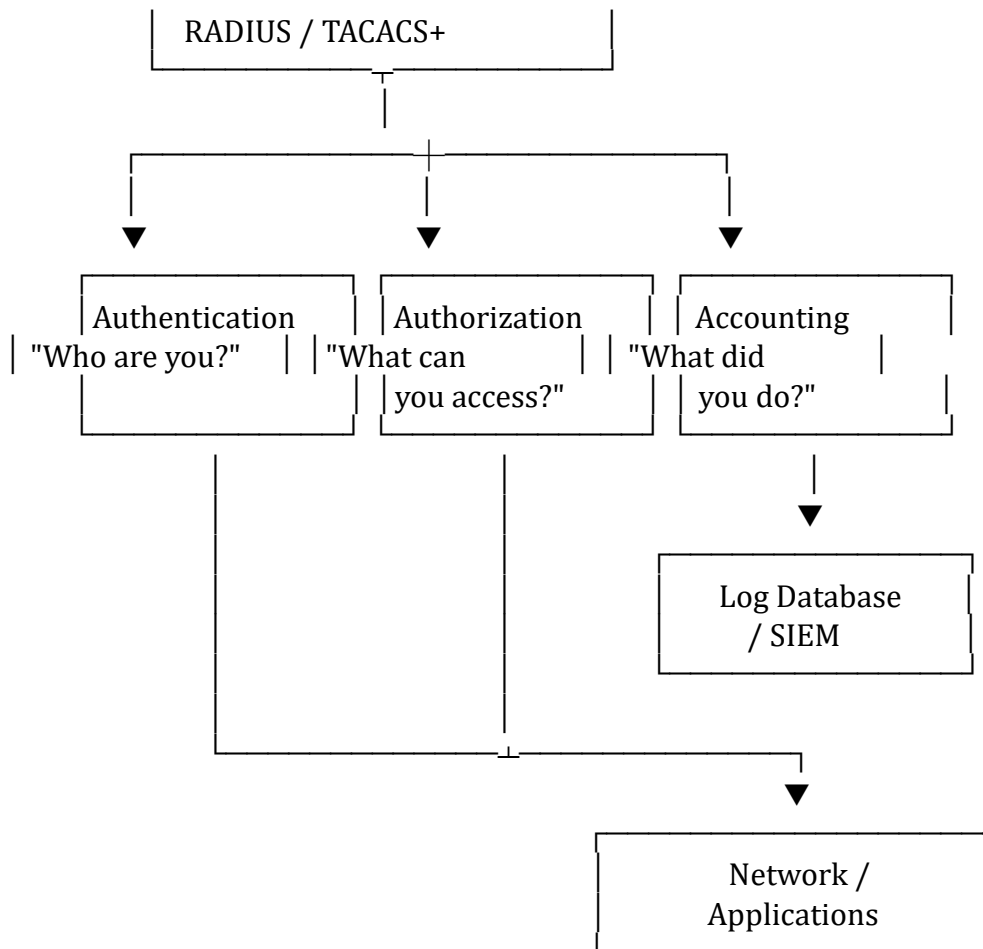
A centralized AAA model allows security policies and user management to be controlled from one location. A distributed AAA model can provide additional availability and scalability when an organization has multiple offices, network locations, or cloud environments.

[Q-3] Draw a simple network diagram (can be hand-drawn and photographed) showing how AAA servers are integrated into a typical enterprise network for user access control, labeling where Authentication, Authorization, and Accounting processes happen.

(A)

○ **Simple Network Diagram**





○ Explanation

Authentication: The AAA server verifies the identity of the user using credentials such as a username and password, certificate, or multi-factor authentication.

Authorization: After authentication, the AAA server determines the user's permissions and decides which VLANs, network resources, applications, or services the user can access.

Accounting: The AAA system records login and logout times, IP addresses, session information, access attempts, and other relevant activity for auditing and security purposes.

[Q-4] Pick any app you use daily (Instagram, Paytm, or Flipkart) and write a paragraph explaining how AAA concepts could be applied to secure user logins and actions within that app. Hint: Think about what happens when you log in, what

features you are allowed to use, and how your actions might be tracked for security or billing.

(A)

○ AAA in Instagram

Instagram can use AAA concepts to secure user accounts and control access to different features. Authentication occurs when a user logs in using their username, password, email address, phone number, or additional security methods such as two-factor authentication.

After authentication, authorization determines what the user is allowed to do, such as posting content, sending messages, changing account settings, or managing a business account.

Different users and account types may have different permissions. Accounting can record important account and security activities such as login attempts, devices used to access the account, active sessions, password changes, and other security-related actions. These records can help detect suspicious activity and improve account security.

[Q-5] Use ChatGPT or any AI tool to generate a sample AAA policy for a college WiFi network that restricts access to students only, allows certain websites, and logs all access attempts. Paste the generated policy and briefly explain how each AAA function is addressed.

(A)

○ College WiFi AAA Security Policy

1. Authentication Policy

1. Only registered college students are allowed to access the student WiFi network.
2. Each student must use their unique college username and password.
3. Shared accounts are not permitted.

4. The AAA/RADIUS server must verify every login request.
5. Multiple failed login attempts should result in temporary account restrictions or additional security checks.

2. Authorization Policy

1. Successfully authenticated students may access the Internet and approved educational websites.
2. Students may access websites and services such as:
 - College website
 - Learning Management System (LMS)
 - Online educational resources
 - Approved academic portals
3. Access to restricted internal systems, administrative servers, and sensitive resources should be denied.
4. Network access can be assigned according to the student's role or department.
5. Guest users should use a separate guest WiFi network.

3. Accounting Policy

1. All successful and failed login attempts should be recorded.
2. The system should record:
 - Student username
 - Date and time
 - IP address
 - Device information, when available
 - Session duration
 - Successful and failed authentication attempts
3. AAA logs should be stored securely for auditing and security investigations.
4. Suspicious activity and repeated failed login attempts should be reviewed by the network administrator.

○ How the AAA Functions Are Addressed

Authentication: It verifies that the person connecting to the college WiFi is a registered student.

Authorization: It determines which websites, services, network resources, and privileges the authenticated student can use.

Accounting: It records access attempts and session activity so that administrators can monitor network usage, investigate security incidents, and maintain audit records.

○ Overall AAA Flow

